METRIC
MIL-DTL-24673A
28 April 2020
SUPERSEDING
DOD-B-24673
31 July 1986

DETAIL SPECIFICATION

BELLS, SHIPS AND SMALL CRAFT

This specification is approved for use by all Departments and Agencies of the Department of the Defense.

1. SCOPE

1.1 Scope. This specification covers bells for use aboard ships and small craft.

1.2 Classification. Bells furnished under this specification are of the following sizes as specified (see 6.2):

a. Size 1 – 1.4-kilogram bell
b. Size 2 – 2.2-kilogram bell
c. Size 3 – 4.5-kilogram bell
d. Size 4 – 9-kilogram bell
e. Size 5 – 14-kilogram bell
f. Size 6 – 27-kilogram bell
g. Size 7 – 34-kilogram bell
h. Size 8 – 90-kilogram bell

1.3 Part or identifying number (PIN). PINs to be used for bells acquired to this specification are created as follows:

M	**24673**	-	**X**
Prefix for Military Specification	Specification Number	-	Size (1 through 8)
Examples: M24673-1 is the PIN for a 1.4-kilogram bell and M24673-6 is the PIN for a 27-kilogram bell.			

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 6350

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-H28 - Screw-Thread Standards for Federal Services

(Copies of this document are available online at https://quicksearch.dla.mil.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

ASTM INTERNATIONAL

ASTM B584 - Standard Specification for Copper Alloy Sand Castings for General Applications

(Copies of this document are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material.

3.1.1 Bell bodies and clappers. Bell bodies and clappers shall be copper-silicon alloy castings in accordance with ASTM B584, Copper Alloy, UNS No. C87200. Bell castings shall be free from internal stresses or strains that may cause cracking of the bell.

3.1.2 Fittings. Fittings such as bolts, nuts, washers, pins, and supporting lugs shall be of copper-silicon alloy in accordance with ASTM B584.

3.2 Construction. The construction mass and dimensions shall conform to figures 1, 2, 3, and 4. The shape of the bodies and clappers shall be as shown on figures 1, 2, 3, and 4.

3.3 Marking. The letters “U.S.” shall be cast in relief (raised), or engraved no deeper than, 1 millimeter (mm) and shall conform to figures 1, 2, 3, and 4.

3.4 Finish. The exterior of the bell and the interior from the open end of the bell to the area where the clapper strikes shall have a smooth polished finish and be free of any manufacturer or country of origin markings.

3.5 Acoustic properties. Bells shall have full, clear, round, and far-reaching tones as specified in table I when examined as specified in 4.3. Tone key frequency and sound pressure requirements for 1.4- and 2.2-kilogram bells are not required; however, the examination of 4.3 is required.

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TABLE I. Tone keys and sound pressure levels.

Bell mouth diameter (mm)	Bell mass (kg)	Tone key (approx.)	Fundamental frequency (Hz±10)	Sound pressure level (dB) (minimum) re: $2x10^{-5}$ pascal
200	4.5	"A"$_6$ Natural	1760	110
250	9.0	"G"$_6$ Natural	1570	111
300	14.0	"C" ♯$_6$	1100	112
360	27.0	"B"$_5$ Natural	990	114
400	34.0	"A"$_5$ Natural	880	115
550	90.0	"F" ♯$_5$	740	117

3.6 Workmanship. Surfaces, including threads, shall be free from burrs, sharp edges, and adhering sand. Materials shall be sound, of uniform quality and condition, and free from seams, cracks, and other defects that may adversely affect the strength, endurance, usefulness/serviceability, appearance, or wear resistance of the part. Any material that has been treated in a manner to conceal defects shall not be offered for Government acceptance.

3.7 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

4. VERIFICATION

4.1 Conformance inspection. Conformance inspection shall include the test of 4.2 and the examination of 4.3. Each bell shall be supported by the yoke or eyebolt with the clapper in place and rung by striking the side of the bell with the clapper. The clapper shall be activated by a lanyard.

4.2 Acoustic properties. The tone key produced shall be determined by comparison with a tuning fork calibrated to the fundamental frequency of the bell, or by measurement of the actual frequency of the bell sound pressure levels as specified in table I. Table I requirements shall be determined by measurement at a 1-meter distance from the bell.

4.3 Visual examination. While the bell is supported as specified in 4.1, it shall be visibly examined for compliance with the requirements of 3.1.1 through 3.6.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The bells covered by this specification are intended for use as specified in table II. Current stock of bells may be used until depleted; future requisitions for bells should refer to the current version of this specification.

TABLE II. Intended use.

Bell mouth diameter (mm)	Bell mass (kg)	Usage
150 and 200	1.4 and 2.2	Small crafts less than 12 meters in length.
200	4.5	Boats 12 meters and over in length but less than 20 meters in length, and ships watch and fire bell.
250	9.0	Navy and marine craft 12 meters and over in length but less than 20 meters in length. Submarines less than 100 meters in length.
300	14.0	Ships and boats 20 meters and over in length but less than 50 meters. Submarines over 100 meters in length.
360	27.0	Ships 50 meters or over in length but less than 75 meters in length.
400	34.0	Ships 75 meters or over in length but less than 150 meters in length.
550	90.0	Ships 150 meters or over in length.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Size of bell required (see 1.2).

c. Packaging requirements (see 5.1).

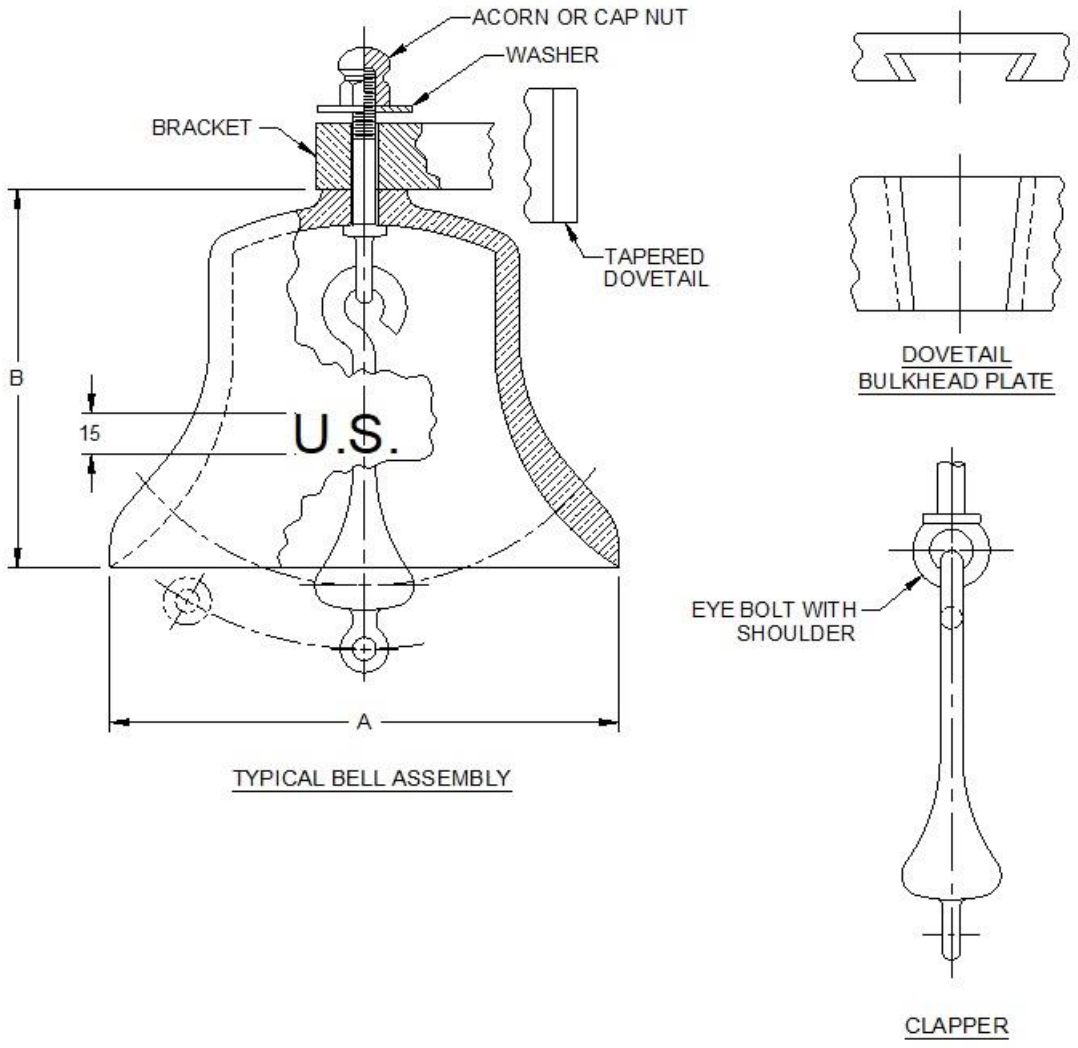
6.3 Subject term (key word) listing.

Castings

Sound pressure

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

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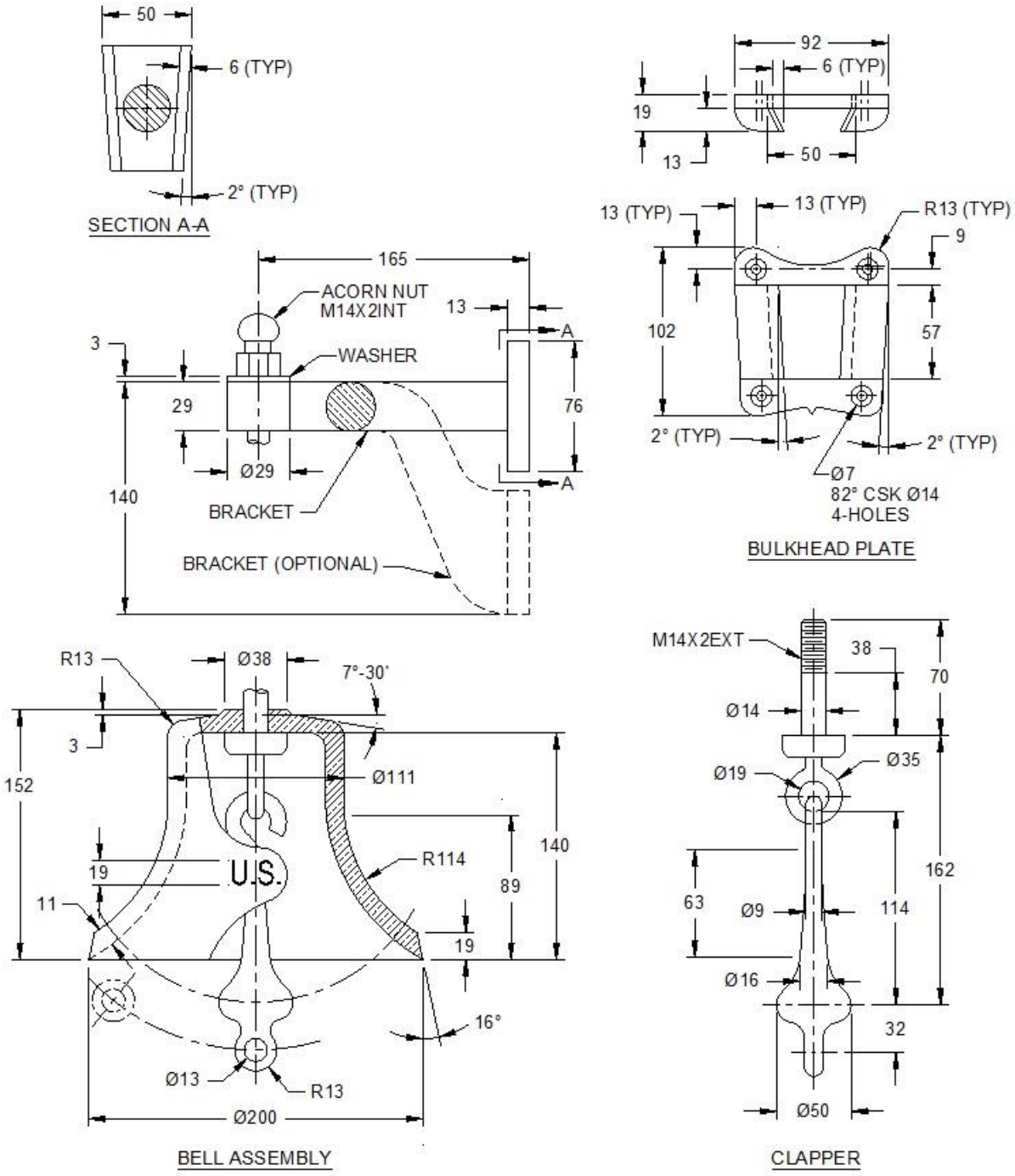
Mass of bell in kilograms	Dimensions in millimeters except as noted	
	A	B
1.4	150	105
2.2	200	133

NOTES:

1. Dimensions in millimeters. Mass of clapper shall be no less than 3 percent of bell mass. Tolerance: Bell mass and clapper ±0.2 kilogram.
2. Except where otherwise indicated, tolerances on dimensions less than 150 mm are ±2 mm; over 150 mm are ±3 mm.
3. Screw threads shall be in accordance with FED-STD-H28.

FIGURE 1. 1.4- and 2.2-kilogram bell.

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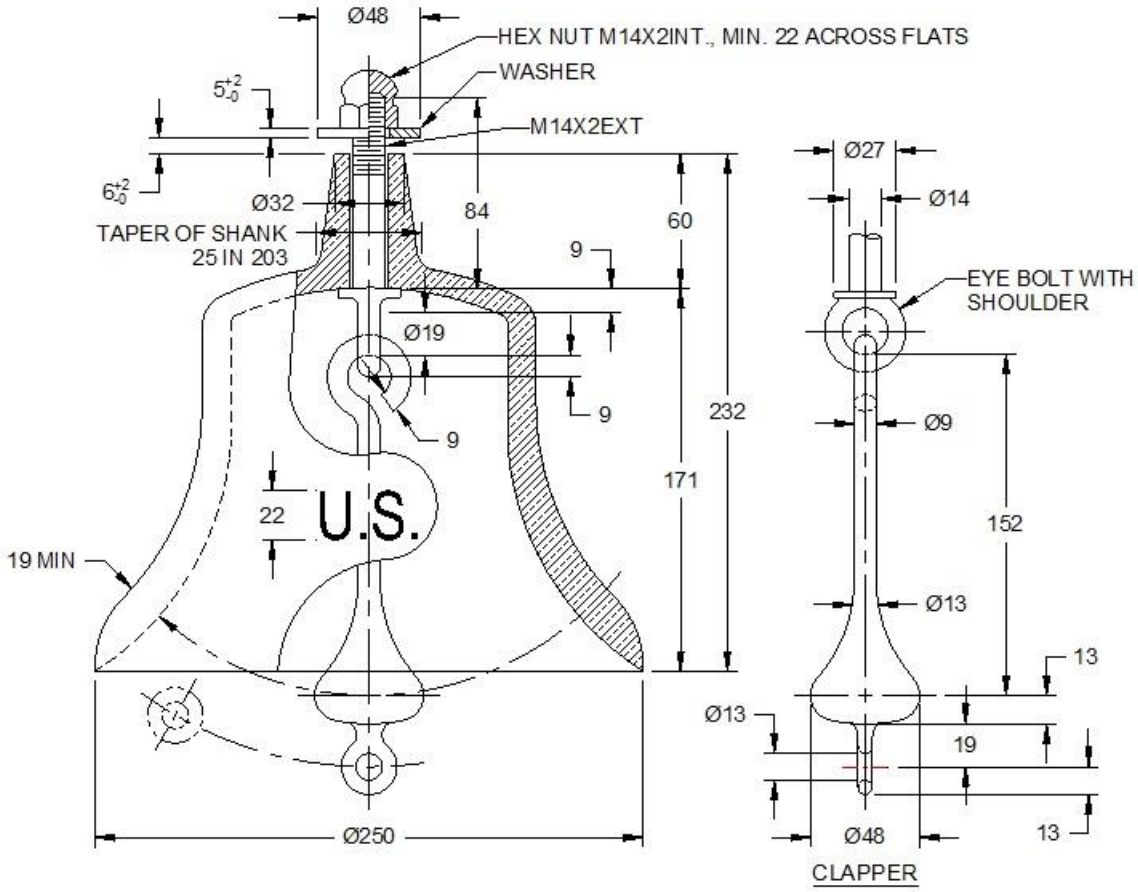


NOTES:

1. Dimensions in millimeters. Mass of clapper shall be no less than 3 percent of bell mass. Tolerance: Bell mass and clapper ±0.3 kilogram.
2. Screw threads shall be in accordance with FED-STD-H28.

FIGURE 2. 4.5-kilogram bell.

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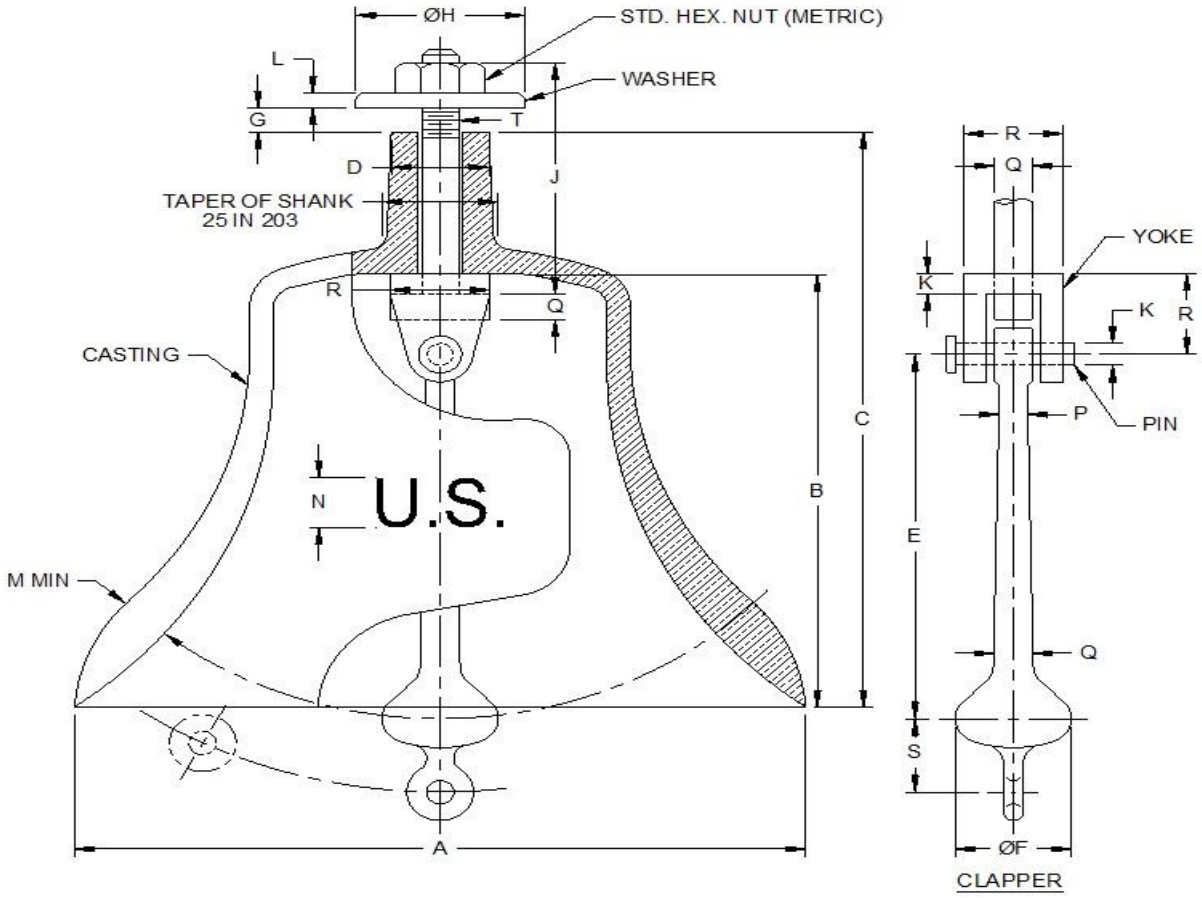


NOTES:

1. Except where otherwise indicated, tolerances on dimensions less than 125 mm are ±2 mm; over 125 mm are ±3 mm.
2. Mass of bell castings: 9.0±0.45 kilograms.
3. Mass of clapper: 0.5±0.05 kilogram.
4. Dimensions in millimeters.
5. Screw threads shall be in accordance with FED-STD-H28.

FIGURE 3. 9-kilogram bell.

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Mass of bell only (kilograms)		Mass of clapper (kilograms)		Dimensions in millimeters except as noted																	
Tolerance	Nominal	Tolerance	Nominal	A	B	C	D	E	F	G	H	J	K	L	M	N	P	Q	R	S	T
14	0.70	0.60	0.04	300	200	267	38	168	48	13	64	108	10	5	21	25	13	16	38	35	M16 x 2
27	1.35	0.95	0.07	360	248	317	51	216	54	14	83	119	11	6	26	29	16	20	44	38	M20 x 2.5
34	1.70	1.36	0.09	400	276	349	57	241	57	16	102	127	13	6	29	29	19	24	51	38	M24 x 3
90	3.60	3.40	0.17	550	375	476	83	333	89	19	127	165	16	10	39	32	22	24	63	51	M24 x 3

Tolerance on dimensions G, K, and L is +2/-0. All other tolerances are as follows:

Dimension	Less than 125	125 to 249	250 to 374	375 to 499	500 to 750
Tolerance	±2	±3	±5	±6	±10

NOTE: Screw threads shall be in accordance with FED-STD-H28.

FIGURE 4. 14- to 90-kilogram bell.

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CONCLUDING MATERIAL

Custodians:
Army – CR4
Navy – SH

Preparing activity:
Navy – SH
(Project 6350-2019-001)

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